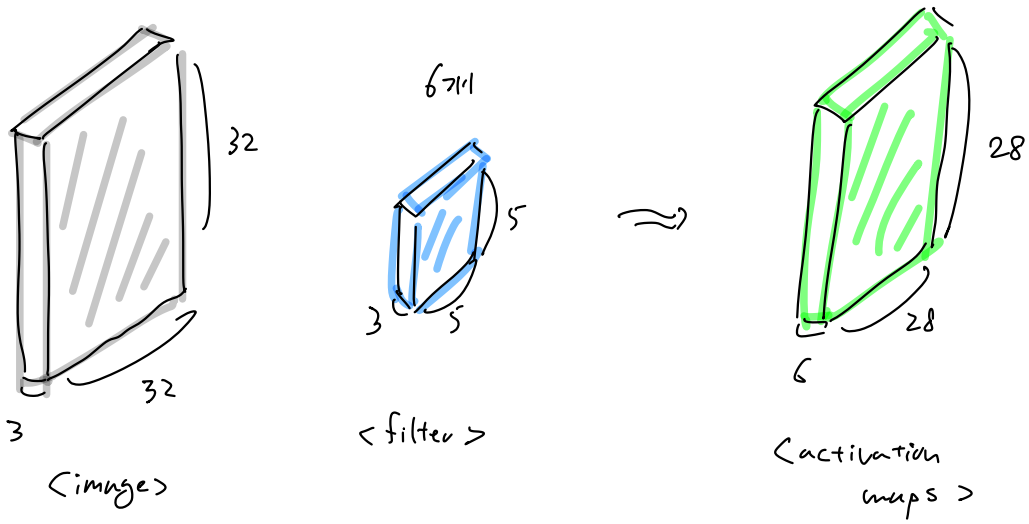
< Convolution Layer >



Naive Conv

\+ { padding, stride }

input : $N \times N \times C$

filter : $K \times K \times C$ × D개 ⇒ output $N' \times N' \times C'$

$S(N'-1) = N - K + 2P$

필터가 $(N'-1)$번을 S만큼 이동한 거리

= 필터가 실제로 이동가능한 공간의 크기

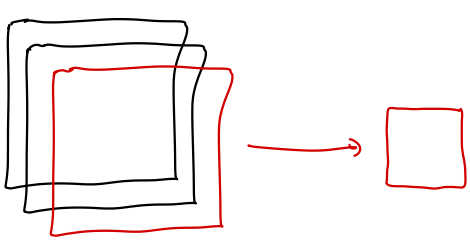
$C' = D$

1x1 Conv Layers

$56 \times 56 \times 64$

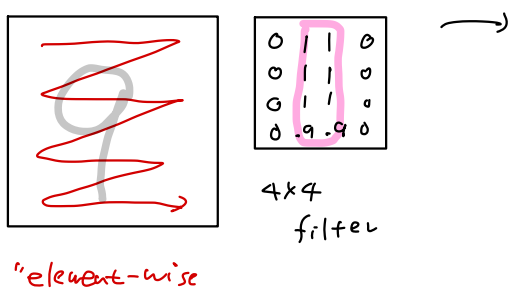
$1 \times 1 \times 64$ 32개 ⇒ $56 \times 56 \times 32$

Pooling layer



- max sampling
- average sampling

필터: filter



"element-wise multiplications & average"

∀ locations

translation invariance (이동 불변성)

Convolution은 translation에 대해 equivariance를 가진다.

$$f \circ g(x) = g \circ f(x)$$

Conv trans trans Conv

AlexNet

227×227×3	Input
55×55×96	Conv 1 (kernel = 11×11, stride = 4, pad = 0) + ReLU
27×27×96	Max Pool 1 (kernel = 3×3, stride = 2)
27×27×96	~~Norm 1~~
27×27×256	Conv 2 (kernel = 5×5, stride = 1, pad = 2) + ReLU
13×13×256	Max Pool 2 (kernel = 3×3, stride = 2)
13×13×256	~~Norm 2~~
13×13×384	Conv 3 (kernel = 3×3, stride = 1, pad = 1) + ReLU
13×13×384	Conv 4 (kernel = 3×3, stride = 1, pad = 1) + ReLU
13×13×256	Conv 5 (kernel = 3×3, stride = 1, pad = 1) + ReLU
6×6×256	Max Pool 3 (kernel = 3×3, stride = 2)
[4096]	FC6
[4096]	FC7
[1000]	FC8

① ReLU ⇒ saturating gradient ⇓

② first stride ⇑ ⇒ feature size & computation ⇓

③ Normalization is not used in recent CNNs

④ FC dropout 50% ⇒ overfitting ⇓

cf. dropout

신경망의 일부분 무작위로 비활성화

- co-adaptation ⇓ "뉴런들끼리 data가 뭐였는지"
- ensemble ⇑

ZFNet

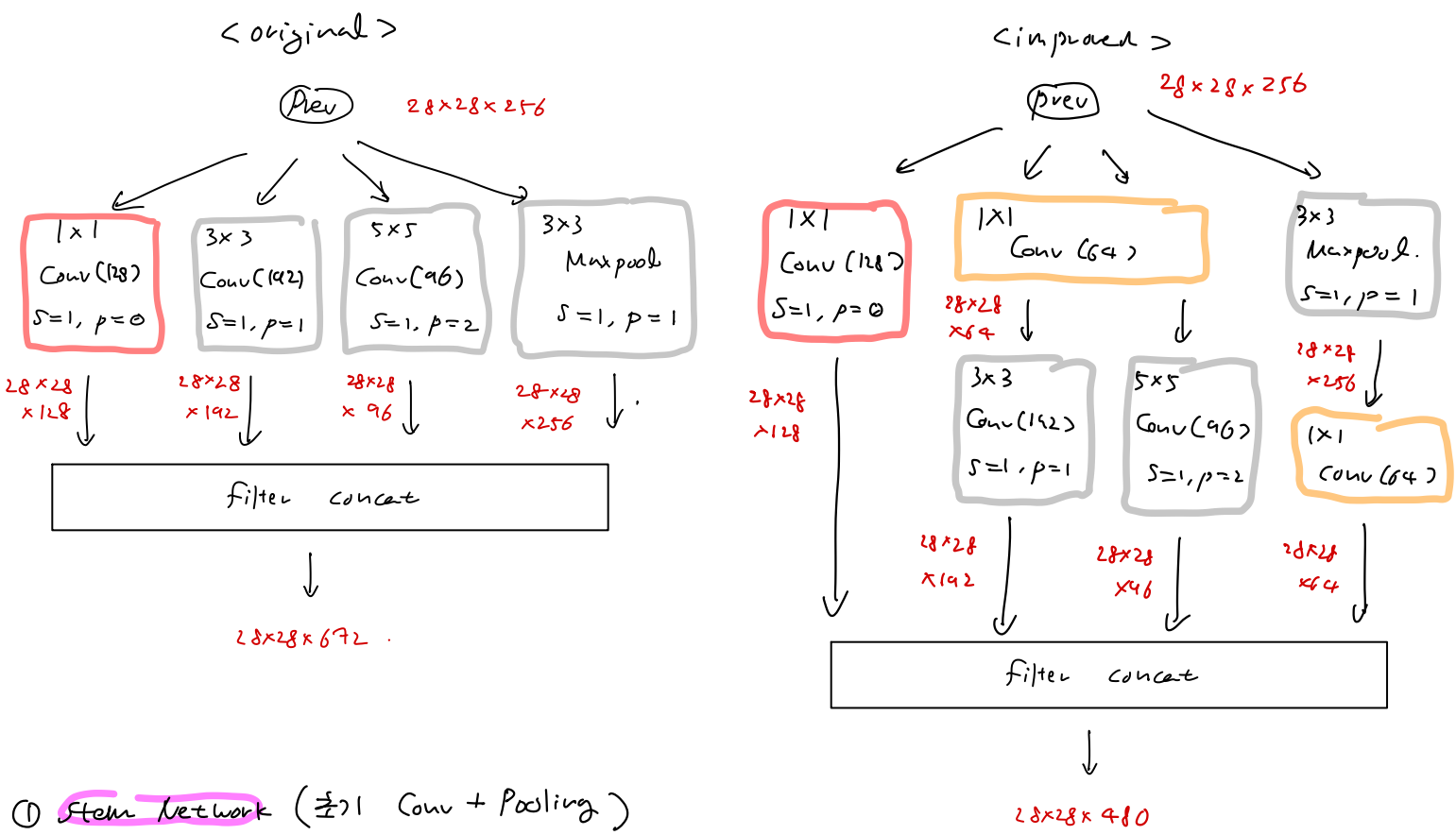
- Conv1 (~~11×11~~ stride ~~4~~)
  7×7 2

- Conv3, 4, 5 : ~~384~~, ~~384~~, ~~256~~ filters
  512 1024 512

GoogleNet

< Inception Modules >

parameter size (filter) ⇓



① Stem Network (초기 Conv + Pooling)

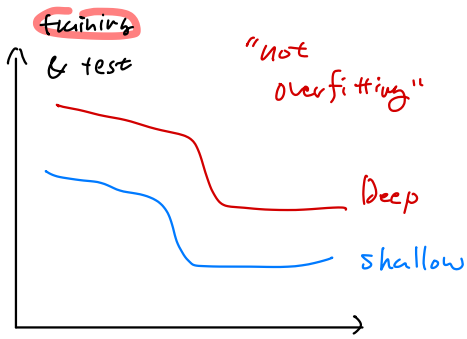
② Inception Module Stack

③ Classifier

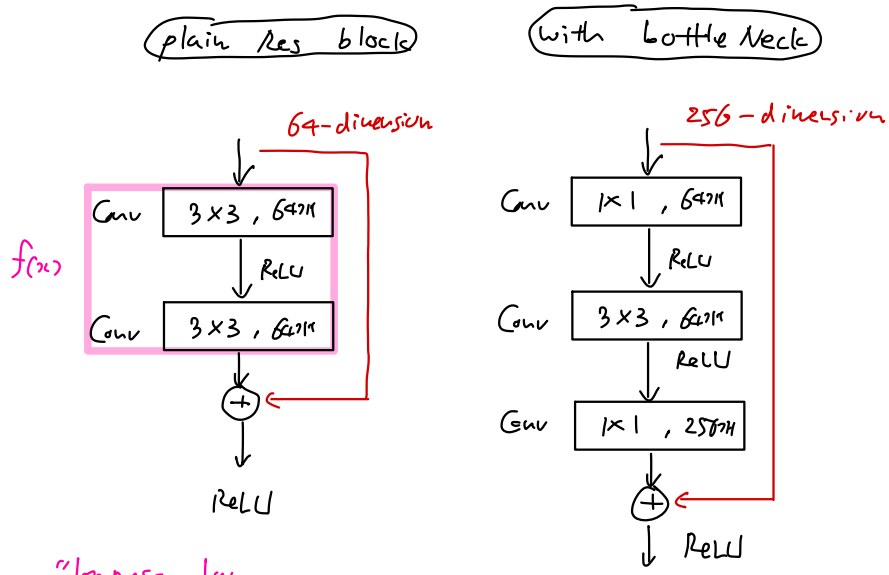
(i) 7×7×1024 —average pooling→ 1×1×1024

(ii) Auxiliary Classifiers : 중간의 노드 분류기에 그래디언트 주입

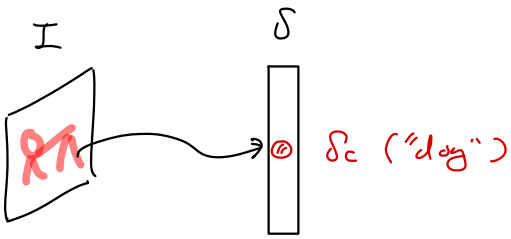
Deeper Network?



< Residual Connection >



"bypass layers by setting $f(x) = 0$"



Saliency

$$\frac{\partial S_c}{\partial I} = \frac{\partial F^{(L)}}{\partial F^{(L-1)}} \cdot \frac{\partial F^{(L-1)}}{\partial F^{(L-2)}} \cdots \frac{\partial F^{(1)}}{\partial I}.$$